TraCast

Traffic-speed forecasting with recurrent encoders, road-distance graphs, and Flower

ivanimmanuel-dev / TraCast | Research brief - October 2026

Research question

How much do recurrent encoder choice, published sensor relationships, and federated model averaging change traffic forecasts under controlled training budgets?

Centralized benchmark

Model	MAE (SD)	RMSE (SD)	Parameters
Persistence	4.100	8.549	0
GRU	3.409 (0.032)	7.179 (0.033)	11,136
LSTM	3.363 (0.007)	7.091 (0.031)	12,736
Graph-GRU	3.364 (0.036)	7.107 (0.106)	15,760
Graph-GRU, self only	3.368 (0.023)	7.073 (0.047)	15,760

Errors in mph; three training seeds on one fixed test period. Persistence is deterministic and evaluated once.

LSTM and Graph-GRU reached nearly equal mean MAE. Graph-GRU improved over GRU by 0.0453 mph, while removing graph neighbors changed mean MAE by only 0.0043 mph. The edge effect changed sign across seeds.

Data and evaluation

METR-LA contains 34,272 five-minute timestamps and 207 sensors. The benchmark uses 16 topology-selected sensors across the full available date range, March 1-June 27, 2012: 548,352 speed entries before masking.

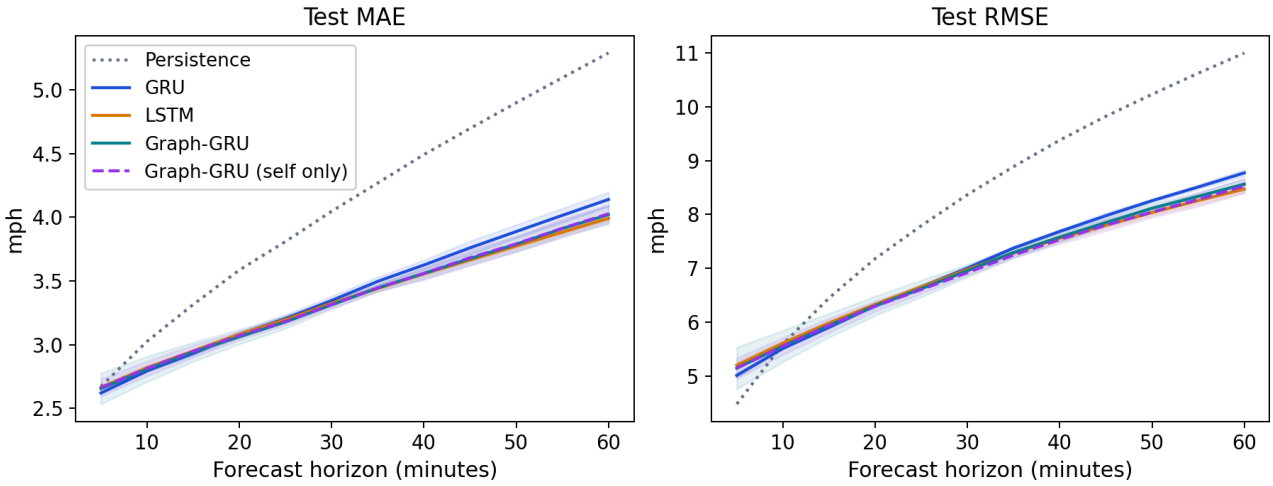
Rows split chronologically 70/10/20 before making sequences. Twelve input steps predict the next twelve steps. Means and scales use observed training values only. Inputs forward-fill within their split; zero/NaN targets remain masked. Validation MAE selects checkpoints.

Graph-GRU architecture



The graph comes from the DCRNN author's distance table. Its Gaussian kernel, sensor IDs, and file hashes are recorded. The selected graph has 85 directed non-self relationships; no speed correlation or held-out targets determine its edges.

Forecast horizon errors



Federated forecasting

The expanded Flower study used 12 sensors, 14 days, 3 clients, and 3 seeds. At five rounds, balanced partitions scored MAE 3.774 mph versus 3.750 for the matched pooled reference. Congestion-grouped clients scored 3.836 mph. Training histogram divergence was 0.0169 versus 0.0307 bits. The five-round parameter payload was 721,152 bytes. Round budgets of 1/3/5 and all per-seed results are recorded.

The federated task shares a univariate GRU across sensors. Local training-only scalers, masks, time boundaries, and initial weights match its pooled reference. Model updates use sample-count weighting; errors pool by valid target count. These client partitions are simulated on one host, with no differential privacy or secure aggregation.

Execution and reproducibility

Five epochs per learned model, seeds 7/42/2026, batch size 64, hidden width 32. Training updates averaged 11.1 s for GRU, 10.9 s for LSTM, and 12.7 s for Graph-GRU. Whole-process peak memory was about 320 MiB on a Ryzen 7 5825U with four PyTorch threads.

Configs, source/data hashes, normalization statistics, convergence logs, horizon metrics, timings, model hashes, and regression tests accompany the reports. Historical baseline results remain unchanged. Commands fetch checksum-verified inputs and use pinned Python/PyTorch/Flower dependencies.

Interpretation

The encoder comparison favors LSTM's smaller parameter count at nearly the same mean error as Graph-GRU. The graph ablation shows a small, mixed seed-level edge contribution. Three seeds on one test block describe initialization variation, not uncertainty across future traffic periods. Communication estimates cover tensor values; runtime overhead and real-network costs are separate.

Code and evidence

github.com/ivanimmanuel-dev/TraCast
Full methods, raw metric tables, topology provenance, experiment configurations, and figures: reports/research-report.md and DATA.md.

References: Li et al., ICLR 2018 (METR-LA/DCRNN); Cho et al., EMNLP 2014 (GRU); Hochreiter and Schmidhuber, Neural Computation 1997 (LSTM); McMahan et al., AISTATS 2017 (FedAvg); Flower 1.42.0.